

M Rajesh

Software Engineer | 5.4 Years experience

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Summary

Experienced Software Engineer with 5.4 years of expertise in backend engineering, microservices architecture, and AI-powered application development, with a strong specialization in Large Language Models (LLMs), agentic workflows, and intelligent automation systems. Skilled in designing and building scalable AI solutions using Python, FastAPI, LangChain, Strands Agents, MCP (Model Context Protocol), and vector databases, with hands-on experience in Retrieval-Augmented Generation (RAG), tool orchestration, memory-driven agents, and model-agnostic agent design. Proven expertise in developing and deploying production-grade LLM applications on AWS using ECS Fargate, Bedrock, Docker, and CI/CD pipelines. Passionate about advancing AI innovation through intelligent agent systems, multi-agent orchestration, and next-generation autonomous workflows.

Skills

Python

OOP, FastAPI, Design Patterns and Principles.

AWS

AWS Bedrock, ECS Fargate, AWS Lambda, API Gateway, S3, CDK, and Serverless Architecture.

Langchain & Strands Agents

Agent framework, MCP and tool orchestration, model-agnostic agents, Prompt Engineering, Memory & Chains, Vector Databases, Retrieval-Augmented Generation (RAG).

Docker

Docker file, Volume mounts, and Docker Compose.

Professional Experience

Apr 2024 – present

Konovo, Software Engineer - Product

- Worked as a **Backend Engineer**, developing and optimizing AI-powered chatbot and **agentic solutions** using Python, Strands Agents, AWS services, and modern LLM frameworks.
- Built and orchestrated agentic workflows involving multi-step reasoning, **tool orchestration**, memory handling, and model-agnostic agent design to support complex AI use cases.
- Developed and deployed **MCP** (Model Context Protocol) servers using **FastMCP**, enabling standardized tool interoperability and seamless integration of external tools into agentic ecosystems.
- **Integrated MCP servers** into agentic workflows, allowing agents to dynamically discover, invoke, and coordinate multiple tools across distributed AI systems.
- Played a key role in **system architecture design**, ensuring scalability, efficiency, modularity, and seamless integration of AI, backend, and cloud-native components.
- Collaborated closely with **stakeholders** to gather requirements, define system enhancements, and align AI agent capabilities with business objectives.
- **Containerized** FastAPI, StrandsAgents, and MCP-based applications using **Docker** for streamlined deployment, portability, and scalability.

- Implemented **CI/CD pipelines** using GitHub and AWS CDK for automated build, testing, and deployment of LLM and agentic services across multiple environments.
- Deployed containerized LLM and AI agent applications on **AWS ECS (Fargate)** with blue-green deployments, enabling zero-downtime releases and safe rollbacks in production
- Designed and managed **auto-scaling strategies** on ECS Fargate to handle variable AI/LLM traffic while optimizing performance, availability, and infrastructure cost.

Jul 2022 – Mar 2024
Bangalore

Datma India Pvt Ltd, Software Engineer - Product

Backend Developer

- Led backend development for the **Electronic Health Record (EHR)** project, creating and refining APIs for smooth data handling.
- Developed a Java-based **Keycloak extension** to monitor user activities like logins and admin actions such as user management, enhancing system transparency.
- Created a **Python SDK** to simplify authorization and data access in Datma services, reducing complexity and improving system efficiency.
- Built an **audit logging library** in Go lang to track data transactions in line with FHIR standards, ensuring data security and compliance.

Sep 2020 – Jun 2022
Bangalore

i-exceed technologies solutions private limited, Software Engineer - Product

Appzillon web-IDE Project (IDE for Appzillon developers)

- Worked as a Java (Spring Boot) **Backend Developer**
- Worked on the conversion of an existing monolith application to **microservices** (Using Spring Boot)
- Successfully worked at an independent level while created all the APIs required for **web-ide** (Rest API's)
- Handled the **API security** with keycloak and **spring security**
- Worked with **Apache CouchDB** for DB requests (NoSQL DB)
- Integrated **GIT as a feature** for managing all projects in web-ide (Version Control)
- Created an **API gateway** for routing (spring-cloud-gateway)
- Created a **Discovery server** for all other services to register (Netflix Eureka)

Awards

Employee of the Month

Awarded "Employee of the Month" twice in recognition of outstanding contributions to projects and team success.

Best Team

Recognized as part of the "Best Team" for two consecutive quarters, demonstrating exceptional teamwork, collaboration, and project delivery.

Most Promising New Comer

Presented for demonstrating the continued commitment required to achieve excellence and success.

Education

Aug 2016 – May 2020
Bangalore

Bachelor of Engineering in Computer Science,
Nagarjuna college of engineering and technology

Tools

- Git
- Jira
- VS Code
- Postman
- GitHub Copilot

Projects

Quiz Hub [🔗](https://thequizhub.netlify.app/), <https://thequizhub.netlify.app/>

AI Quiz Generator – Built a full-stack quiz app using React, Python (FastAPI), LangChain, and PostgreSQL. Leveraged OpenAI APIs via LangChain to dynamically generate topic-based quizzes with difficulty and question control. Implemented Google OAuth and structured backend logic with database storage.